

# Ultra SuperServer SYS-120U-TNR

1U Ultra with 12 NVMe/SAS/SATA bays



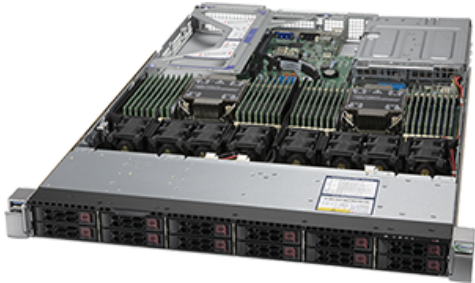
More details here

## Key Applications

Virtualization, HPC, Cloud Computing, High End Enterprise Server, Software Defined Storage, Application Tier Service Provider, 5G/Telco,

## Key Features

- Dual Socket P+ (LGA-4189) 3rd Gen Intel® Xeon® Scalable Processors;
- Intel® C621A Chipset;
- 32 DIMM Slots; Up to 8TB DRAM; Up to 8TB Intel® Optane™ Persistent Memory (up to 12TB with DRAM); 3200/2933/2666 ECC DDR4 LRDIMM; RDIMM; Intel® Optane™ Persistent Memory 200 series;
- 2 PCIe 4.0 x16 (FH, 10.5"L)  
1 PCIe 4.0 x16 (LP)  
1 Internal PCIe 4.0 x16;
- Flexible networking options;
- 12x2.5" hot-swap hybrid NVMe/SATA/SAS drive bays;
- 8 heavy duty fans with optimal fan speed control;
- 1200W redundant Titanium level power supplies;



|                          |                                                                                                                                                                                                                                                                                          |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Form Factor              | 1U Rackmount<br>Enclosure: 437 x 43 x 739mm (17.2" x 1.7" x 29.1")<br>Package: 605 x 203 x 950mm (23.81" x 7.99" x 37.4")                                                                                                                                                                |
| Processor                | Dual Socket P+ (LGA-4189)<br>3rd Gen Intel® Xeon® Scalable processors<br>Up to 40C/80T; Up to 60MB Cache per CPU                                                                                                                                                                         |
| GPU                      | Max GPU Count: Up to 2 double-width or 2 single-width GPUs<br>Supported GPU: NVIDIA PCIe: A100, A16, A2, L4, T4                                                                                                                                                                          |
| System Memory            | Slot Count: 32 DIMM slots<br>Max Memory (2DPC): Up to 8TB 3200MT/s ECC DDR4 RDIMM/LRDIMM<br>Supports Intel® Optane™ persistent memory 200 series                                                                                                                                         |
| Drive Bays Configuration | Default: Total 12 bays <ul style="list-style-type: none"> <li>• 12 front hot-swap 2.5" NVMe/SAS*/SATA drive bays</li> </ul> (*SAS support may require additional storage controller and/or cables)                                                                                       |
| Expansion Slots          | Default <ul style="list-style-type: none"> <li>• 1 PCIe 4.0 x16 LP slot</li> <li>• 1 PCIe 4.0 x16 LP slot</li> <li>• 2 PCIe 4.0 x16 FH/10.5"L slots</li> </ul>                                                                                                                           |
| On-Board Devices         | SATA: SATA (6Gbps) ; RAID 0/1/5/10 support<br>NVMe: NVMe; RAID 0/1/5/10 support<br>Network Connectivity: 2 RJ45 10GBASE-T with Intel® X710-AT2 (optional)<br>2 RJ45 10GBASE-T with Intel® X710-TM4 (optional)<br>2 SFP+ 10GbE with Intel® X710-TM4 (optional)<br>No NIC option supported |
| Input / Output           | LAN: 1 RJ45 1 GbE Dedicated BMC LAN port<br>USB: 2 USB 3.0 ports(rear)<br>1 USB 3.0 port(front)<br>Video: 1 VGA port<br>DOM: 2 SuperDOM (Disk on Module) ports                                                                                                                           |

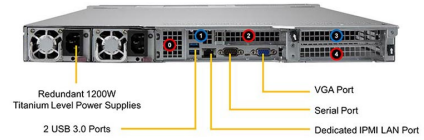
(Front View – System)



| Slot   | Description                                                 |
|--------|-------------------------------------------------------------|
| 1 – 6  | 2.5" Hot-swap NVMe/SAS3/SATA3 Drive Bays (NVMe from CPU1)** |
| 7 – 12 | 2.5" Hot-swap NVMe/SAS3/SATA3 Drive Bays (NVMe from CPU2)** |

\*SAS3 support requires additional parts in optional parts list  
 \*\*NVMe support requires additional parts in optional parts list

(Rear View – System)



| Slot | Description                      |
|------|----------------------------------|
| 1    | Ultra Riser Networking Slot      |
| 2    | PCI-E 4.0 x16 Slot (Internal LP) |
| 3    | PCI-E 4.0 x16 Slot (LP)          |
| 4    | PCI-E 4.0 x16 Slot (FH, 10.5" L) |
| 5    | PCI-E 4.0 x16 Slot (FH, 10.5" L) |

CPU1 CPU2

|                       |                                                                                                                                                                                                                                                                                                                            |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| System Cooling        | Fans: Up to 8x 4cm heavy duty fans with optimal fan speed control<br>Air Shroud: 2 Air Shrouds                                                                                                                                                                                                                             |
| Power Supply          | 1x 1200W Redundant Titanium Level (96%) Hot-plug power supply                                                                                                                                                                                                                                                              |
| System BIOS           | BIOS Type: AMI 32MB SPI Flash EEPROM                                                                                                                                                                                                                                                                                       |
| Management            | SuperCloud Composer®; Supermicro Server Manager (SSM); Supermicro Update Manager (SUM); Supermicro SuperDoctor® 5 (SD5); Super Diagnostics Offline (SDO); Supermicro Thin-Agent Service (TAS); SuperServer Automation Assistant (SAA) New!                                                                                 |
| PC Health Monitoring  | CPU: 7 +1 Phase-switching voltage regulator<br>Monitors for CPU Cores, Chipset Voltages, Memory<br>FAN: Fans with tachometer monitoring<br>Pulse Width Modulated (PWM) fan connectors<br>Status monitor for speed control<br>Temperature: Monitoring for CPU and chassis environment<br>Thermal Control for fan connectors |
| Dimensions and Weight | Weight: Gross Weight: 41 lbs (18.6 kg)<br>Net Weight: 26.5 lbs (12.2 kg)<br>Available Color: Black Front & Silver Body                                                                                                                                                                                                     |
| Operating Environment | Operating Temperature: 10°C to 35°C (50°F to 95°F)<br>Non-operating Temperature: -40°C to 60°C (-40°F to 140°F)<br>Operating Relative Humidity: 8% to 90% (non-condensing)<br>Non-operating Relative Humidity: 5% to 95% (non-condensing)                                                                                  |
| Motherboard           | <a href="#">Super X12DPU-6</a>                                                                                                                                                                                                                                                                                             |
| Chassis               | CSE-119UH3TS-R1K22P-T                                                                                                                                                                                                                                                                                                      |